TidyTuesday Week 45

Pranay Gundam

Tuesday 3rd November, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we throw two unrelated economic series into a blender and act shocked when something comes out. This week's menu: goods-producing employment in Springfield, Massachusetts versus a nursing home price index, and IT manufacturing orders versus the return on equity for Missouri banks (a series so discontinued it has "DISCONTINUED" shouting at us from inside its own title). Truly the frontier of science.

Monday, November 2, gave us the kind of raw regression that makes a layperson gasp and a statistician roll their eyes. Springfield's goods-producing jobs "explained" a jaw-dropping 78 percent of the variation in nursing home prices, with a p-value so small it has to use scientific notation to fit on the page. Astonishing! Groundbreaking! Someone call a journal! And then we detrended it, and the R-squared cratered to a majestic 2 percent with a p-value of 0.52, which in human terms means "we have absolutely nothing here." The entire relationship was two lines quietly drifting in the same direction across the decades, holding hands only because time was passing. This is the house special, folks: a trend mirage, served piping hot and immediately refuted.

Tuesday, November 3, was slightly less embarrassing, which around here counts as a triumph. IT manufacturing orders versus Missouri bank returns started with a modest raw R-squared of 13 percent, and after detrending it dropped to about 6 percent but, plot twist, stayed statistically significant at the 0.01 level. So this one actually survived having its shared trend ripped out from under it, which makes it the rarer and genuinely more interesting kind of result. Do I have a theory? Sure: when the economy is booming, people buy more computers AND banks make more money, so both wiggle together even without the long-run drift. Is that theory worth the electrons it's printed on? Absolutely not, but it holds up better than most of what we find.

As always, the mandatory disclaimer that I will now deliver in a monotone: these series were paired at random, nobody is claiming Springfield's factory workers set nursing home prices or that IT orders run Missouri's banks, and spurious correlation is not a bug of this project but its entire reason for existing. One mirage debunked, one coincidence that refused to die on cue. Call it a productive week.

2 Date: 2026-11-02

Series ID: SMU25781000600000001
This series is titled All Employees: Goods Producing in Springfield, MA-CT (NECTA) and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1990-01-01 and the observation end date is 2024-12-01.The popularity of this series is 1.

Series ID: PRGONUPIHCSA
This series is titled Medical Services Expenditures by Provider: Nursing Homes: Proprietary and Government Nursing Homes Price Index and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2000-01-01 and the observation end date is 2021-01-01.The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_PRGONUPIHCSA	**R-squared:**	0.783
Model:	OLS	**Adj. R-squared:**	0.772
Method:	Least Squares	**F-statistic:**	72.01
Date:	Mon, 02 Nov 2026	**Prob (F-statistic):**	4.64e-08
Time:	21:06:51	**Log-Likelihood:**	-73.979
No. Observations:	22	**AIC:**	152.0
Df Residuals:	20	**BIC:**	154.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	186.2043	11.787	15.797	0.000	161.616	210.793
value_fred_SMU25781000600000001	-2.2652	0.267	-8.486	0.000	-2.822	-1.708

Omnibus:	5.822	**Durbin-Watson:**	0.251
Prob(Omnibus):	0.054	**Jarque-Bera (JB):**	4.103
Skew:	1.043	**Prob(JB):**	0.129
Kurtosis:	3.353	**Cond. No.**	333.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

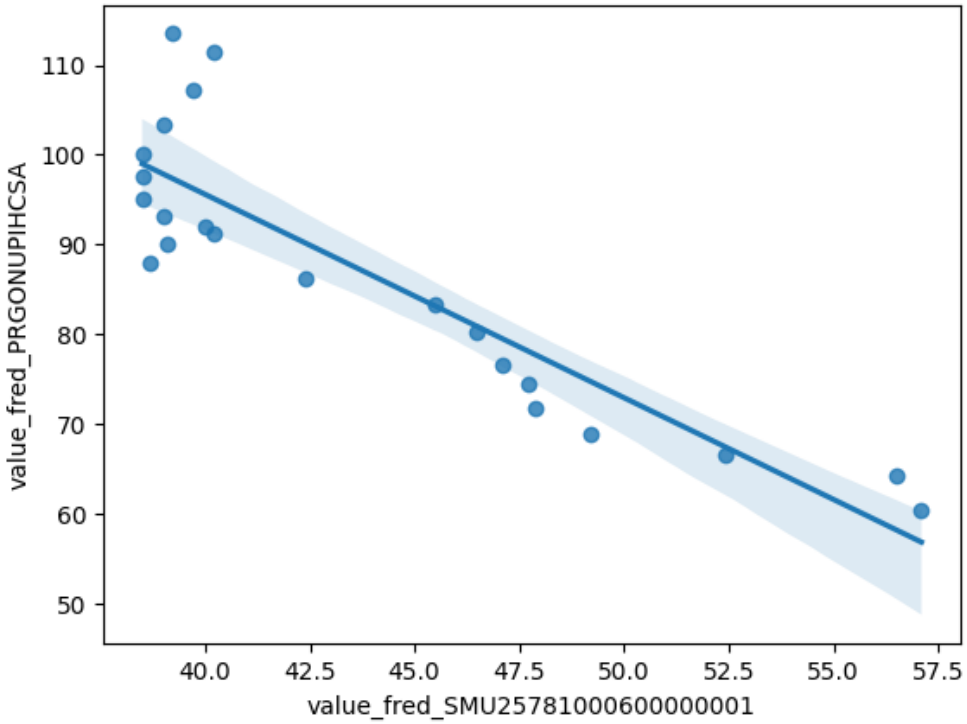


Figure 1: Raw Regression Plot for 2026-11-02

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_PRGONUPIHCSA_detrended	**R-squared:**	0.021
Model:	OLS	**Adj. R-squared:**	-0.028
Method:	Least Squares	**F-statistic:**	0.4223
Date:	Mon, 02 Nov 2026	**Prob (F-statistic):**	0.523
Time:	21:06:51	**Log-Likelihood:**	-42.096
No. Observations:	22	**AIC:**	88.19
Df Residuals:	20	**BIC:**	90.37
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.027e-13	0.367	1.1e-12	1.000	-0.765	0.765
value_fred_SMU25781000600000001_detrended	-0.0866	0.133	-0.650	0.523	-0.365	0.191

Omnibus:	2.262	**Durbin-Watson:**	0.331
Prob(Omnibus):	0.323	**Jarque-Bera (JB):**	1.135
Skew:	-0.097	**Prob(JB):**	0.567
Kurtosis:	1.904	**Cond. No.**	2.75

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

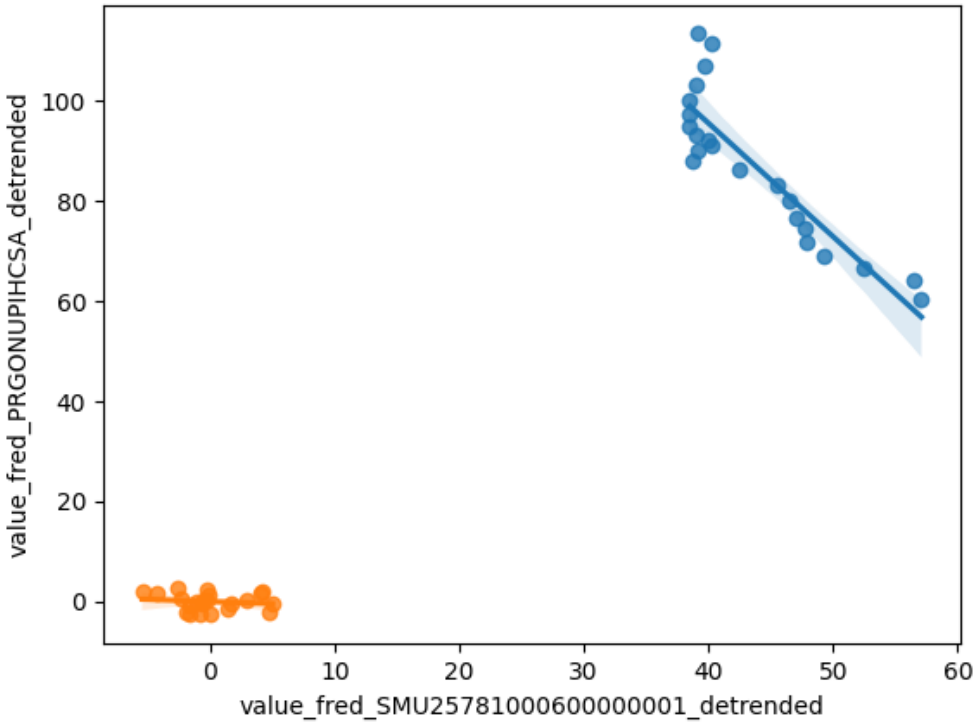


Figure 2: Detrended Regression Plot for 2026-11-02

3 Date: 2026-11-03

Series ID: AITINO
This series is titled Manufacturers' New Orders: Information Technology Industries and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted.The observation start date is 1992-02-01 and the observation end date is 2026-09-01.The popularity of this series is 14.

Series ID: MOROE
This series is titled Return on Average Equity for Banks in Missouri (DISCONTINUED) and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1984-01-01 and the observation end date is 2020-07-01.The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MOROE	**R-squared:**	0.135
Model:	OLS	**Adj. R-squared:**	0.128
Method:	Least Squares	**F-statistic:**	17.51
Date:	Tue, 03 Nov 2026	**Prob (F-statistic):**	5.69e-05
Time:	18:56:19	**Log-Likelihood:**	-307.15
No. Observations:	114	**AIC:**	618.3
Df Residuals:	112	**BIC:**	623.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.9234	2.332	0.825	0.411	-2.697	6.544
value_fred_AITINO	0.0004	9.45e-05	4.185	0.000	0.000	0.001

Omnibus:	41.602	**Durbin-Watson:**	0.161
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	88.948
Skew:	-1.485	**Prob(JB):**	4.84e-20
Kurtosis:	6.147	**Cond. No.**	1.70e+05

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.7e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

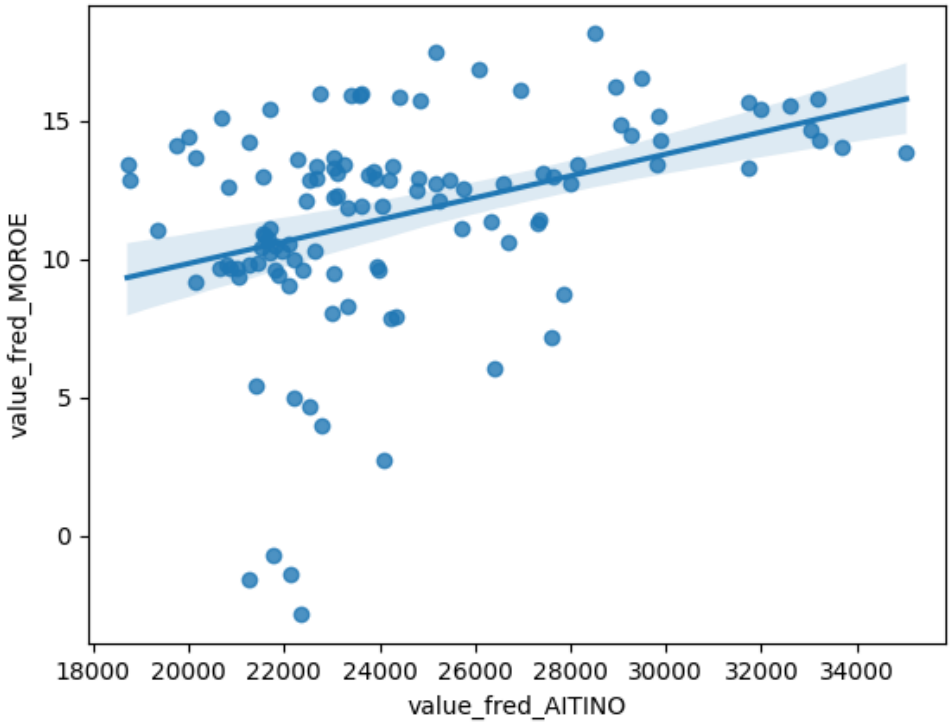


Figure 3: Raw Regression Plot for 2026-11-03

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MOROE_detrended	**R-squared:**	0.057
Model:	OLS	**Adj. R-squared:**	0.049
Method:	Least Squares	**F-statistic:**	6.779
Date:	Tue, 03 Nov 2026	**Prob (F-statistic):**	0.0105
Time:	18:56:19	**Log-Likelihood:**	-295.86
No. Observations:	114	**AIC:**	595.7
Df Residuals:	112	**BIC:**	601.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-7.674e-13	0.306	-2.5e-12	1.000	-0.607	0.607
value_fred_AITINO_detrended	0.0002	9.13e-05	2.604	0.010	5.68e-05	0.000

Omnibus:	60.278	**Durbin-Watson:**	0.190
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	192.530
Skew:	-1.974	**Prob(JB):**	1.56e-42
Kurtosis:	7.994	**Cond. No.**	3.35e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.35e+03. This might indicate that there are strong multicollinearity or other numerical problems.

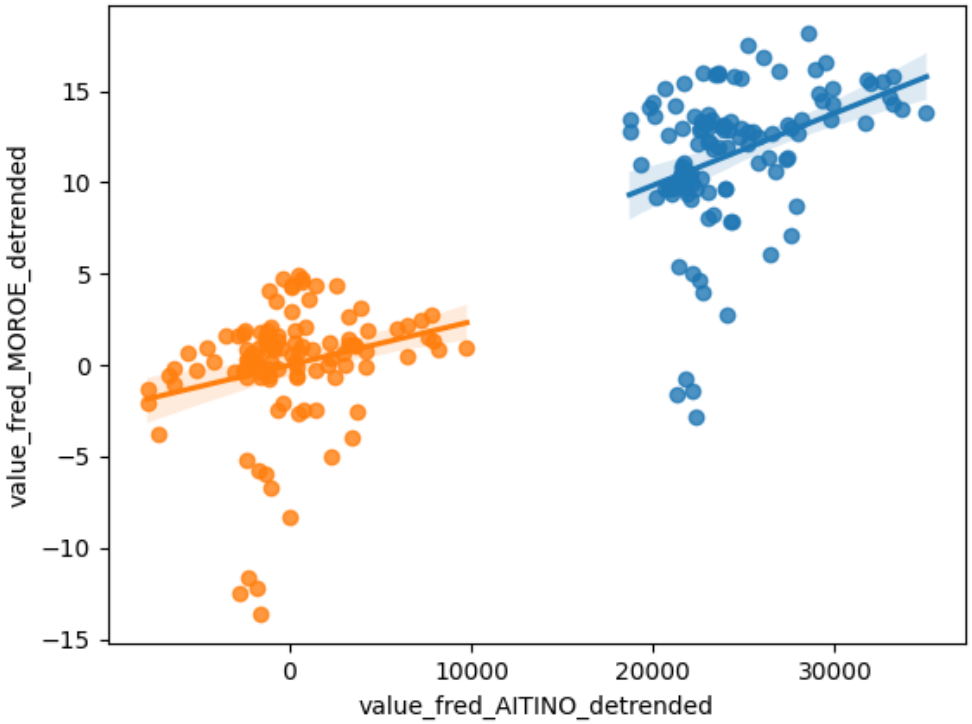


Figure 4: Detrended Regression Plot for 2026-11-03